

LAMBDA, REFERENCE, AND THE SPATIOTEMPORAL DESCRIPTIVE FRAME

Self/Other Polarity, Residual Will, Compression, Translation, and the No-First-Position Result

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EmergenceByDesign

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Related program: Lambda Principle / frame-marked reasoning / Constructive Truth-State / Dimensional Ladder

STANDING | Scope, provenance, and verifier jurisdiction

This paper records independent research and synthesis. It makes no claim of ownership, priority, invention, or discovery over established mathematics, formal logic, Lean, theorem proving, metrology, physics, linguistics, philosophy, information theory, or the cited traditions and results. Established work remains attributable to its sources. The distinctive contribution lies in the proposed organization of reference, self/other polarity, criterion selection, exclusion, compression, spatiotemporal framing, translation, and descriptive residue under the Lambda vocabulary, together with an explicit claim-to-Lean translation layer. No claim requires scientific realism, anti-realism, idealism, materialism, a theology, a theory of consciousness, or a theory of libertarian free will as an undeclared premise. Lean is used here as a checker for encoded propositions, not as a semantic oracle for claims that were never supplied to its kernel.

Abstract

This paper develops a frame-marked account of description around one minimal operational pressure: reference cannot do discriminatory work without preserving a distinction between a source-role and a target-role. The paper calls these poles, at the declared interpretive bridge, the self-side and other-side of reference. It then studies what follows when a criterion partitions a domain, an exclusion is formed relative to that criterion, a comparator operates across the resulting distinction, and a successor description turns a previous boundary or selector into a new referent. The formal core proves role polarity under the stated encoding, nontrivial binary partition, criterion-relative exclusion, observation-fibre obstruction, failure of target factorization under collapsed distinctions, exact-but-noninjective compression, residual selection, non-unique criterion determination, conditional obstruction to absolute descriptive priority, local-versus-universal warrant separation, public-label meaning loss, compositional bridge preservation, possible noncommutation of translations, finite-budget tradeoff, and interpretation variance under a frozen input.

The spatiotemporal application is deliberately conditional. Spatial-type structure is read as a canonical realization of circumscription, localization, separation, and simultaneous contrast; temporal-type structure is read as a canonical realization of succession, traversal, persistence, and directed order. Under explicit bridge premises that a claim of space-first priority uses a temporal-type resource to state priority and that a claim of time-first priority uses a spatial-type resource to objectify or locate its referent, the formal kernel proves a No-First-Position result: neither direction qualifies as absolute descriptive priority under a definition that requires the complementary resource to have been sequestered. The proof does not claim that physical space and physical time literally generate one another.

The paper then turns to compression and language. Successful notation and shared language can condense chains of operations into primitive-like interfaces, conserving attention while hiding construction history. That compression is neither rejected nor demoted. The Lambda discipline asks what a compression preserves, what its fibres identify, what bridge licenses a stronger interpretation, and where residual selection remains. A prior Lean-certified semantic-deflation case supplies a concrete demonstration: the checker can validate distinct formal meanings and their incompatibility while a public label suppresses which meaning was used. The machine does not create the conflation; the reporting projection does. The final design direction treats human-machine collaboration as a cooperative descriptive technology for exposing such bridges and residue, not as a view from nowhere, a privileged benevolent source, or a malicious other.

CENTRAL CALIBRATION

Lean can provide very strong assurance about a proposition that has been encoded and accepted by its kernel. That assurance should be neither weakened nor silently promoted. Kernel acceptance is evidence for the encoded proposition in the executed environment; it is not, by itself, evidence that an informal sentence, a physical identification, a cultural interpretation, or a claim of frame-independent jurisdiction was encoded correctly. The proper response is not to ask Lean to do less. It is to expose the translation boundary so Lean can do exactly what it does well.

Status key

FORMAL RESULT

A proposition derived from the definitions or hypotheses stated in the formal model. Where a Lean

declaration is supplied, the translation ledger names it exactly.

BRIDGE

A declared mapping between formal structure and an informal, linguistic, experiential, or physical reading. A bridge can be argued for and attacked; it does not inherit proof status merely because its source theorem is proved.

INTERPRETIVE PROPOSITION

A disciplined reading of already-fixed structure. It is not presented as a theorem about observer-independent reality.

EMPIRICAL HYPOTHESIS

A claim about actual people, cognition, institutions, communication, attention, or physical behavior that requires evidence beyond logic alone.

DESIGN DIRECTION

A constructive proposal about how descriptive or human-machine systems might be built. It states an objective or architecture, not a discovered law of nature.

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1. Verification has jurisdiction - and translation has a receipt

The motivating problem is not a defect in formal proof. It is a mismatch of jurisdictions. A formal checker receives an expression in a formal language, elaborates it against declarations and definitions, and checks the resulting term against a type-theoretic kernel. The Lean documentation describes the trusted kernel as a small type checker for Lean's core theory. That strength is precisely why a semantic bridge should not be smuggled into its receipt: if the bridge never entered the formal object, kernel acceptance did not adjudicate it [9].

The companion work *Before the Proof* expresses the same discipline as a registration chain: target relation, registration, problem-forming bridge, formal or empirical traversal, scoped result, and interpretive bridge. Verification of the traversal does not automatically verify either bridge [6]. The present paper therefore places its translation receipt before the larger philosophical synthesis. The reader can see which condensed claims have executable formal counterparts and which claims remain bridge statements, interpretations, empirical hypotheses, or design directions.

This ordering matters. A paper can contain a theorem and then immediately use the theorem's vocabulary in a stronger physical or ontological sentence. The proximity of the sentences does not make them one proposition. Likewise, a public badge saying “Lean-verified” may accurately describe a declaration while leaving open whether the declaration formalizes the sentence a reader thinks it formalizes. The proper calibration separates three receipts: translation, kernel execution, and downstream interpretation.

THREE-RECEIPT DISCIPLINE

1) Translation receipt: identifies the informal condensate, its formal declaration, its hypotheses, and what the translation intentionally leaves out. 2) Kernel execution receipt: records the exact source, environment, diagnostics, and accepted declarations from an actual run. 3) Bridge receipt: records the argument by which a proved formal proposition is applied to a linguistic, experiential, or physical target. None substitutes for the others.

This edition supplies the translation ledger and a captured Lean Web execution receipt. Lean 4.33.0 accepted all 23 theorem declarations and the checked supporting definition `SelectedCriterion`, with zero captured errors or warnings. Each of the 23 axiom reports records no axiom dependencies. Appendix B includes the complete final message transcript and the one source correction required by the initial run. The accepted source accompanies the receipt inside this PDF. The companion receipts [4,5] establish the earlier workflow; this edition records an execution of its own source.

2. Formal universe and arrow discipline

The paper distinguishes at least four kinds of arrows. A logical implication is written $\Rightarrow$. A definitional or representational map is written $\rightarrow$ when its source and target are declared. A translation bridge is written $\mapsto$ or named explicitly. Residual selection is described in prose or by the symbol W only when a predecessor registry does not uniquely force the selected continuation. A physical causal arrow is never inferred merely because an equation has been written in a computational direction.

This discipline follows the earlier measurement paper, which separates temporal order from causal order and treats spatial and temporal description as mutually supporting conditions of measurable relations rather than a hierarchy of ontological priority [3]. It also follows the Nested Transfer paper, where a forward mathematical arrow and an inverse reconstruction can both be legitimate without either alone establishing ontological priority [7].

FORMAL BOUNDARY

The formal kernel uses types, predicates, functions, products, Booleans, natural-number inequalities, and explicit hypotheses. Words such as self, other, space, time, will, absolute, and meaning acquire only the roles encoded in the source. The paper argues separately for the intended bridges.

3. Condensate decomposition: the Lean-addressable kernel

The following ledger breaks every formal condensate used as a theorem-level step in this paper into a Lean declaration. The ledger deliberately refuses to “prove by vocabulary.” For example, Lean does not prove that all natural-language reference must be modeled by the RoleTagged construction; it proves what follows once that construction is adopted. Lean does not prove that physical temporal priority necessarily has the bridge property used in the No-First-Position theorem; the bridge property appears as a field of SpatiotemporalBridge and is therefore visible as a premise.

ID	Condensed formal claim	Lean declaration	Boundary
K01	Source-role and target-role are distinct constructors.	<code>source_target_roles_distinct</code>	Pure formal role distinction.
K02	A source-tagged carrier cannot equal a target-tagged carrier.	<code>reference_polarity</code>	Formalization of referential polarity after roles are declared.
K03	Self-reference can reuse one carrier without erasing source/target role difference.	<code>reflexive_reference_preserves_role_distinction</code>	Does not claim a theory of consciousness.
K04	Indexing the same reflexive reference at a later station preserves role polarity.	<code>iterated_reference_polarity</code>	Models relocation across registered stations.
K05	A nontrivial Boolean criterion has at least one true-side and one false-side witness.	<code>nontrivial_criterion_exhibits_poles</code>	Binary partition kernel.
K06	A nontrivial Boolean criterion therefore has at least two distinct source states.	<code>nontrivial_criterion_exhibits_distinct_states</code>	Formal distinction, not physical distance.
K07	Exclusion by a criterion is the criterion-relative negation.	<code>exclusion_is_criterion_relative</code>	Does not define every philosophical use of “metaphysical.”
K08	Deterministic postprocessing cannot split a projection fibre already identified.	<code>postprocessing_cannot_split_fibre</code>	Lambda observation-fibre core.

K09	If two projected-equal states differ on a target, that target cannot factor through the projection.	target_difference_blocks_factorization	Standard factorization obstruction.
K10	A compression can be exact for one declared target while remaining non-injective on its source.	target_exact_compression_can Lose_other_distinctions	Formal pharmakon/compression exemplar.
K11	Two admissible procedures with different outcomes block a unique outcome from the registered conditions.	residual_selection	Residual Selection Theorem skeleton.
K12	A chosen admissible procedure in such a non-unique registry satisfies the formal ResidualWill predicate.	residual_will_from_competing_admissible_procedures	Preserves the established project definition of will.
K13	If two admissible procedures induce different criterion maps, no unique criterion map is forced by those conditions.	competing_admissible_criteria_not_uniquely_forced	Formalizes criterion-selection residue.
K14	If every priority claim uses a complementary resource, no claim can satisfy priority plus non-use of that resource.	complement_requirement_blocks_absolute_priority	Generic absolute-priority obstruction.
K15	Given the two explicit spatiotemporal bridge premises, neither space-first nor time-first satisfies absolute descriptive priority.	no_first_position	Conditional theorem; physical applicability rests on bridge arguments.
K16	A proposition may have warrant in one frame while universal warrant across the frame class fails.	local_warrant_not_universal_in_general	Countermodel to automatic promotion.
K17	Merely naming a predicate Absolute does not supply a universal elimination rule Absolute $p \rightarrow p$.	absolute_label_does_not_self_license	Does not deny that “absolute” is syntactically meaningful.
K18	Two usage modes can expose the same public label.	same_public_label	Surface-label identity only.
K19	The same public label can coincide while the operative meanings differ on a witness.	same_label_can_hide_different_meanings	Compact semantic-deflation model.
K20	Predicate preservation composes across two declared translation maps.	predicate_preservation_composes	Positive bridge theorem.
K21	Three individually defined translation maps need not commute.	translations_need_not_commute	Local mappings do not imply global path-independence.
K22	If full requested resource demand exceeds a finite budget, both demands cannot simultaneously receive their full requested allocations under that budget.	finite_budget_forces_tradeoff	Resource theorem only; human attention requires an empirical bridge.
K23	One frozen input can receive different outputs under different interpretation functions.	same_input_can_receive_different_interpretations	Countermodel to “same inscription implies same interpretation.”

BRIDGE LEDGER - NOT CLAIMED AS KERNEL CONSEQUENCES

B01: meaningful natural-language reference is modeled by a source-role/target-role relation. B02: the

source-role is read as the self-side and the target-role as the other-side of the current reference. B03: spatial description is read as a canonical realization of localization/circumscription; temporal description as a canonical realization of directed order/traversal. B04: the relevant “space first” causal-priority claims use directed temporal-type order, and the relevant “time first” objectification claims use localization/circumscription. B05: actual human immersion and attention allocation instantiate the finite-budget model. B06: human-machine contrast and translation practices improve communicative performance. These are argued, tested, or proposed separately; the Lean ledger does not silently certify them.

4. Translation receipt and execution boundary

LEAN EXECUTION RECEIPT - PASS

Artifact: LambdaSpatiotemporalFrame.lean
 Environment: Lean Web / mathlib-stable / Lean 4.33.0; import Lean only
 Accepted: 23 theorem checks + SelectedCriterion; 19 supporting definitions
 Diagnostics: 0 errors; 0 warnings; 49 messages accounted for
 Axiom audit: 23/23 reports contain no axiom dependencies
 Source identity: exact accepted-editor match; 13,340 UTF-8 bytes
 Capture: 10 September 2026, 22:48:56 UTC
 Full transcript, source correction, and reproduction instructions: Appendix B.

The first execution exposed one missing type annotation in K06. The conclusion $\exists x y : X, x \neq y$ now declares the witness domain explicitly. The full corrected file then passed. The final receipt records all 24 declaration checks, 23 axiom reports, the version string, and the completion marker.

The prior digest has been removed. Source identity rests on exact comparison with the accepted editor text, preserved as the embedded Lean file and the captured source-bearing Lean Web link. A later change to that source requires another execution. Editing the PDF around an unchanged source does not change the propositions Lean checked.

The translation ledger records the intended correspondence between prose and formal declarations. The execution receipt records acceptance of those declarations with their explicit parameters and hypotheses. The bridge ledger retains its own jurisdiction; an empty axiom-dependency report does not discharge the stated bridge premises.

CALIBRATION RULE

Lean acceptance $\Rightarrow$ the encoded declaration was accepted in the reported execution environment. Lean acceptance does NOT, without another encoded argument, imply InformalStatementMatch,

PhysicalIdentification, OntologicalExhaustion, CommunityAcceptance, or FrameIndependentJurisdiction. Conversely, lack of such stronger status does not weaken the accepted formal result inside its actual scope.

5. Reference polarity: source, target, and self/other

The paper begins below personhood. It does not define self as a biological organism, conscious ego, soul, or legal individual. It begins with the operational structure of reference. A referential description distinguishes a role from which reference is made and a role occupied by what is referred to. In the Lean model, those roles are constructors `RefRole.source` and `RefRole.target`. A carrier tagged as source cannot equal a carrier tagged as target because the roles differ (K01-K02).

The substantive bridge is then declared: “self-side” names the source-side of the current referential distinction; “other-side” names its target-side. The terminology does not claim that the two sides are substances or that the source is a human consciousness. It marks the minimal polarity required for the reference relation to discriminate a target. If no difference survives between the roles, the chosen representation no longer carries the directional information by which one side refers to the other.

$\text{MeaningfulRef}(\text{source}, \text{target}) \mapsto \text{DistinctRole}(\text{source-side}, \text{target-side})$

BRIDGE - SELF/OTHER

Self/other is used here as a role-polarity of reference. The paper does not infer that observer-independent reality was “first divided” into metaphysical substances called Self and Other. It says that a referential frame cannot usefully erase the distinction by which its reference is oriented and still claim to preserve that referential information.

This framing also answers the apparent counterexample of an impersonal mathematical model. A coordinate origin, a selected variable, a domain element, or a theorem statement may not contain a person. Yet using any of them as a referent still occurs within a descriptive act whose source-side and target-side can be typed even if ordinary notation suppresses the source index. The source-side may be inherited from a community, formal environment, compiler, paper, speaker, or current model context. Omitting the label “self” does not demonstrate that the referential polarity has been eliminated; it may only demonstrate that the polarity has been compressed.

6. Reflexive reference: folding is not erasure

Self-reference supplies the harder test. Suppose the same carrier x occupies both roles. The formal source represents the two occurrences as `asSource x` and `asTarget x` and proves they remain unequal as role-tagged objects (K03). This does not claim that x has become two substances. It claims that the act “ x refers to x ” still contains two relational positions. Reusing the carrier folds the relation back on itself; it does not collapse the source-role into the target-role while preserving the same directed reference.

$\text{carrier}(x) = \text{carrier}(x), \text{ but } \text{role}(\text{source}) \neq \text{role}(\text{target})$

The indexed theorem K04 repeats the same fact at a registered station n . This supplies the minimal formal kernel for recursive description: a prior source, criterion, selector, or frame can become the target of a successor act of reference, but that successor act again has a source-side and a target-side. The previous boundary may become content. The act of making it content does not, by itself, create an unindexed exterior from which all reference can be viewed at once.

FORMAL RESULT - REFLEXIVE REFERENCE BOUNDARY

Within the role-tagged encoding, self-reference and indexed repetition preserve source/target role distinction. The interpretive consequence is modest but important: objectifying the prior source relocates the current reference boundary; it does not prove that reference as such has been escaped.

7. Criterion, partition, complement, and exclusion

A binary partition requires a criterion. Let $c : X \rightarrow \text{Bool}$. The formal predicates `SelfSide c x` and `OtherSide c x` record the true and false sides. A `NontrivialCriterion` requires at least one witness on each side, and K05-K06 prove both poles exist and cannot be represented by the same state. This is not a claim that all meaning is Boolean. It isolates the minimum structure needed when a binary distinction is actually invoked.

$$c : X \rightarrow \{\text{true}, \text{false}\}; \quad \text{Nontrivial}(c) = \exists x, y \text{ with } c(x)=\text{true} \text{ and } c(y)=\text{false}$$

Exclusion then has an explicit dependency. In the formal kernel `ExcludedBy P x` is defined as $\neg P x$; K07 verifies that the excluded side is criterion-relative. The important descriptive point is that the excluded condition has not disappeared. It remains present negatively through the predicate whose complement defines the exclusion. To call something “nonphysical,” for example, in a usage where that word means not satisfying a declared physical predicate, requires the physical predicate to have been fixed well enough for its negation to be formed.

INTERPRETIVE PROPOSITION - EXCLUSION CARRIES ITS BOUNDARY

An exclusion is informative only relative to a criterion, domain, and logical operation that make the excluded class intelligible. This does not settle every historical or philosophical use of “metaphysical.” It blocks the narrower move in which an excluded category is treated as though it arrived without the partition that made the exclusion possible.

8. Residual selection and the established meaning of will

The term `will` is not introduced anew here. The companion `Constructive Truth-State` paper reserves `will` for residual selection not already forced by the registered conditions [2]. The present Lean kernel preserves that definition. `UniqueOutcome` says that every admissible procedure yields one common outcome. K11 proves that if two admissible procedures yield different outcomes, `UniqueOutcome` fails. `ResidualWill` then requires that a chosen procedure is admissible and that unique outcome is not forced by the current registry; K12 derives that condition from competing admissible procedures.

CANONICAL PROJECT DEFINITION - WILL

`Will` names the residual act or distinction of selection not already forced by the registered predecessor conditions. The term does not, by itself, attribute the selection to personal agency, consciousness, libertarian freedom, randomness, or an uncaused power. A mechanism may later explain a previously residual selection in an enlarged registry; that successor explanation does not retroactively erase the earlier scope statement.

The criterion question now becomes exact. A family `criterionOf : J → X → Bool` lets different admissible procedures induce different partitions. K13 proves that if two admissible procedures induce

different criterion functions, then the predecessor conditions do not uniquely determine one criterion. A selected procedure can therefore index the operative criterion without the paper pretending that the criterion was already forced by the previous account.

underdetermined registry $\leadsto$ W / selected procedure $\rightarrow$ criterion $\rightarrow$ partition $\rightarrow$ comparator

The first arrow is deliberately not labeled “cause.” It marks residual selection relative to a declared account. Once the criterion is fixed, ordinary deterministic operations may follow. If a later model identifies a mechanism that uniquely fixes the selector, the residual moves upstream or disappears at that enlarged level. The vocabulary therefore functions as an accounting discipline rather than a commitment to an agential metaphysics.

9. Lambda: projection, fibres, and non-sequestration

The Lambda principle is used here as a registry name for a recurring structural condition already instantiated in the Dimensional Ladder: a representation may remain exact for its declared target while identifying distinctions that matter to another target; deterministic processing of the compressed representation cannot reconstruct a distinction already collapsed inside one of its fibres [4]. K08 formalizes the elementary postprocessing result. K09 strengthens it to a factorization obstruction: if $\text{projection}(x_1) = \text{projection}(x_2)$ but $\text{target}(x_1) \neq \text{target}(x_2)$, no function defined only on the projected value can reproduce that target on both states.

$$\Phi(x_1) = \Phi(x_2) \text{ and } T(x_1) \neq T(x_2) \Rightarrow T \text{ does not factor through } \Phi$$

K10 supplies the reciprocal positive fact: compression can be completely exact for one declared target while non-injective on its source. This matters because the Lambda claim is not “compression is false.” It is that target adequacy and source exhaustion are different predicates. A one-bit answer may perfectly solve the question it was designed to answer while discarding every source distinction irrelevant to that question.

LAMBDA ENLARGEMENT LAW

A distinction invisible inside the current projection becomes expressible only through an enlargement that supplies an independent relation capable of separating the relevant fibre, or through a theorem establishing that the hidden distinction is irrelevant to the declared target. Projection is not sequestration: absence from the visible projection is not proof of absence from the source relation.

10. Compression, primitives, and the pharmakon of successful notation

Mature notation is efficient because it does not require a user to replay the derivation of every familiar operation. The Dimensional Ladder calls this the pharmakon of familiar operations: a reciprocal rule, projective chart transition, or algebraic identity can become so compressed by fluency that the operation that forced it disappears from attention [4]. The language paper extends the same observation to grammar and public status labels [1].

The cure proposed here is not decompression at all times. That would make communication and calculation unusably expensive. The cure is reversible compression: preserve enough metadata to reopen the path when a dispute, translation, or new target makes a hidden distinction relevant. K10 gives the exact formal core. A map may be target-exact and source-noninjective. The danger enters only when target adequacy is substituted for source identity or ontological exhaustion.

PHARMAKON OF COMPRESSION

Compression is both enabling and potentially concealing. It conserves finite attention and supports shared action. The same success can make selected primitives, default predicates, and bridge assumptions appear unconstructed. The Lambda discipline therefore asks not whether compression should be abolished, but whether its preserved target, erased indices, and reopening path remain inspectable.

11. Spatiotemporal descriptive resources

The spatiotemporal argument becomes strongest when it begins with structural resources rather than with a claim that abstract logic is literally made of physical space and physical time. Define two bridge families. A localization/circumscription resource supports distinction, placement, separation, boundary, adjacency, and the identification of a referential locus. A directed-order/traversal resource supports succession, persistence, precedence, transition, ordered comparison, and the directionality required by many causal-priority claims.

Physical spatial description is then treated as a canonical realization of the first family, and physical temporal description as a canonical realization of the second. The earlier Increment, Registry, and Measure paper supplies the operational motivation: measured temporal increments require distinguishable change, while measured spatial separation requires reference and an operation by which difference can be registered or compared [3]. Neither direction is promoted there to an ontological first cause.

BRIDGE S - SPATIAL-TYPE RESOURCE

Localization/circumscription/individuation $\mapsto$ spatial-type description when the domain is physical or geometric. The bridge does not identify every logical distinction with literal Euclidean space.

BRIDGE T - TEMPORAL-TYPE RESOURCE

Directed order/traversal/succession $\mapsto$ temporal-type description when the domain is physical or chronological. The bridge does not identify every mathematical relation with physical time. Symmetric or static relations need not carry temporal direction.

This distinction also prevents an easy equivocation around causation. Temporal precedence does not by itself establish causation, and causal models can encode additional domain-specific constraints. The no-first argument therefore applies to causal-priority formulations only when their stated semantics actually require directed antecedence, production, traversal, or another temporal-type ordering resource. A different semantics must be audited on its own terms.

12. The No-First-Position Result

The formal result is intentionally transparent. For predicates Priority and UsesComplement, define AbsolutePriority(c) as Priority(c) together with non-use of the complementary resource. If every Priority claim entails UsesComplement, then no AbsolutePriority witness exists. K14 proves this generic statement. The SpatiotemporalBridge structure then carries two explicit premises: every SpaceFirst claim in the admitted class uses a temporal resource, and every TimeFirst claim uses a spatial resource. K15 proves the conjunction of the two resulting nonexistence claims.

```

AbsolutePriority(P,U,c) := P(c)  $\wedge$   $\neg$ U(c)
SpaceFirst(c)  $\Rightarrow$  UsesTemporalResource(c)
TimeFirst(c)  $\Rightarrow$  UsesSpatialResource(c)
-----
 $\neg\exists c$  AbsoluteSpaceFirst(c)  $\wedge$   $\neg\exists c$  AbsoluteTimeFirst(c)

```

The theorem does not hide where the substantive work lives. Lean can make the consequence of the bridge premises exact. It cannot decide, merely from the English words space, time, cause, first, or fundamental, whether a particular scientific or philosophical sentence satisfies those bridge premises. That classification belongs in the translation and bridge receipts.

12.1 Space-first pressure

Consider a sentence of the form “space causes time,” where causes means that the first term is productively or antecedently prior to the second. To express that priority, the sentence introduces a directed order between distinguishable terms. Under the declared temporal-type bridge, the allegedly derivative resource is already used by the statement of derivation. The result is not that the sentence is meaningless. It is that the sentence does not meet the paper's stronger criterion of absolute descriptive priority, which requires the complementary resource to have been sequestered from the warrant of priority.

12.2 Time-first pressure

Consider the reciprocal claim that time is the independently sufficient first term from which spatiality derives. For time to function as the distinguished subject of that claim, the description must individuate or circumscribe the referent well enough to contrast it with what is attributed to it. Under the declared spatial-type bridge, that objectification uses a localization/distinction resource. Again, the conclusion is not that the sentence cannot be written. It is that its warrant does not qualify as complement-free absolute priority under the stated criterion.

12.3 What the theorem does not say

- It does not prove that physical space and physical time are identical.
- It does not prove that they literally cause one another.
- It does not prove simultaneity, a common beginning, no beginning, cyclic time, or any particular cosmology.
- It does not prohibit local reductions such as $T=f(S)$ or $S=g(T)$ inside a model.
- It does not prohibit a critic from rejecting one of the bridge premises; doing so identifies the exact attack surface rather than dissolving it in vocabulary.

13. “Absolute” as content versus absolute jurisdiction

The paper does not argue that a phrase such as “a conclusion about the absolute” fails syntactically. A formal or natural language can introduce a token Absolute and construct well-formed sentences with it. The issue is the stipulation that would let the token carry the jurisdiction its name suggests. K17 makes exactly this narrow point: there is no theorem of pure logic saying that every predicate someone chooses to call Absolute automatically licenses the elimination rule $\text{Absolute}(p) \Rightarrow p$.

K16 supplies the corresponding frame example. A proposition can have warrant in one declared frame while universal warrant across a class of frames fails. The existence of the local proof does not become weaker because the universal statement fails. The distinction is between the strength of the local warrant and the scope over which that warrant has been established.

FORMAL RESULT - CONTENT DOES NOT SELF-GRANT JURISDICTION

A statement can meaningfully assert universality inside a declared language. Its content may quantify over a large or even universal-looking domain. The act of writing that content does not by itself prove that the interpretation, domain, bridge, and acceptance rule are frame-independent. An invariance or correspondence argument must carry that additional burden.

This gives the paper a deliberately non-skeptical conclusion. Frame-independent truth is neither asserted nor denied by the formal kernel. A realism axiom, correspondence theory, or stronger invariance theorem may be added. What the present result prevents is automatic promotion from “proved here under these commitments” to “this formal act has certified its own independence from every commitment needed to interpret it.”

14. Axioms, defaults, and primitive-like compression

Axioms and primitives make formal work possible by establishing a stopping layer. A user of arithmetic does not re-prove the identity laws every time a number is added. A speaker does not reconstruct a grammar before every sentence. A proof assistant elaborates against an environment of declarations rather than asking the kernel to derive the language from no prior language. These are not defects; they are the infrastructure that makes finite reasoning tractable.

The Lambda question concerns promotion. Operative primitivity does not imply absolute fundamentality. A relation may function as primitive at one station because its construction has been compressed, inherited, or deliberately frozen. A successor inquiry can reopen that primitive as an object and examine the operation by which it was selected. The successor station still has its own primitives, interpretation, and stopping rule. Recursive reopening therefore expands accountability without claiming a final view from nowhere.

INTERPRETIVE PROPOSITION - IMMERSIVE DEFAULT

A successful primitive-like interface can become immersive: the user spends attention on operations performed through the interface rather than on reconstructing the interface itself. This describes a plausible role for linguistic and mathematical fluency. The claim that specific human neural systems implement this exact architecture remains empirical, not a theorem.

15. Finite attention, immersion, and explicit empirical scope

The finite-budget theorem K22 isolates the resource logic without smuggling in psychology. If two tasks require costs costA and costB , the available budget is smaller than their sum, and an allocation remains within budget, then it is impossible for both tasks simultaneously to receive allocations at least as large as their full costs. The proof is elementary natural-number arithmetic.

$$\text{budget} < \text{costA} + \text{costB} \Rightarrow \text{not}(\text{fullA} \wedge \text{fullB} \text{ under that budget})$$

This permits a disciplined bridge to attention. A finite agent may have more potentially relevant distinctions, frames, or reconstruction tasks than can receive maximal processing simultaneously. Compression, heuristics, defaults, and delegated verification can then function as resource-saving strategies. The companion Constructive Truth-State paper explicitly classifies survival allocation and social compression as empirical hypotheses rather than logical consequences [2]. The present paper keeps that classification.

EMPIRICAL HYPOTHESIS - ATTENTIONAL IMMERSION

Actual human cognition often allocates limited attention to a currently operative frame while relying on compressed primitives, learned norms, and proxy checks for background structure. This may help explain why successful framing operations recede from awareness. It requires empirical study; K22 proves only the abstract budget tradeoff.

16. Semantic time and the moving interpretive frame

A frozen inscription need not freeze interpretation. K23 gives the smallest countermodel: two functions can map the same input to different outputs. The Constructive Truth-State paper develops the richer indexed form $M_t = I_t(R_t)$, in which R can remain unchanged while I changes [2]. “Semantic time” names the ordered change of interpretation without claiming that semantic time is a new physical substance.

This matters for recursive self-description. A later frame may reproduce the bytes of an earlier statement, quote it, formalize it, or even reconstruct a prior computational state. The later act still occurs under a successor interpretation unless that interpretation is separately constrained. Additional metadata can narrow the admissible interpretations, but the metadata itself becomes another inscription. The result is not a proof that invariant interpretation is impossible; it is a proof that more inscription, merely because it is more inscription, does not eliminate the need for interpretation.

17. Communication as a partial semantic handshake

Communication among agents does not require their complete descriptive frames to be identical. It requires enough structure to survive translation for the current purpose. Let $f : A \rightarrow B$ translate source expressions or states, and let P and Q be the corresponding predicates. The formal relation $\text{PreservesPredicate}(f, P, Q)$ requires $P(a) \leftrightarrow Q(f(a))$ for every admitted a . K20 proves that if $A \rightarrow B$ and $B \rightarrow C$ each preserve the relevant predicate, their composition preserves it from A to C .

This supports the “checksum of customs” metaphor in a precise, limited way. A greeting, proof notation, unit convention, logical connective, social role, or technical term can function as a compact handshake that tests enough shared expectations to proceed without comparing complete internal histories. The handshake is target-relative. Two agents may agree on the checksum required for one exchange and diverge on another part of their frames.

INTERPRETIVE PROPOSITION - SEMANTIC HANDSHAKE

A shared convention can be modeled as a compact invariance test across a translation bridge. Successful local preservation licenses the current communicative operation; it does not prove that the participants possess identical meanings, histories, values, or ontologies.

This is why cultural normalization can be both stabilizing and concealing. Fluent speakers inherit a large library of handshakes before they can inspect the historical operations that produced them. The language paper treats language as a socially transmitted technology for segmentation, naming, relating, compressing, preserving, and coordinating experience [1]. A second grammar, logical tradition, number base, or formalization can function as parallax: not automatically superior, but capable of exposing what the first representation made transparent through habit.

18. Noncommuting translations and network-scale conflict

K21 supplies a minimal counterexample to global path-independence. Three maps AB, BC, and AC can be perfectly well-defined while $AC(x)$ differs from $BC(AB(x))$. The theorem is tiny because the point is structural: the existence of local translations does not force a commuting global diagram.

$$T_{AC}(x) \text{ need not equal } T_{BC}(T_{AB}(x))$$

At network scale this gives a formal place to locate semantic drift. If a term passes through several communities, institutions, disciplines, or machine systems, each local translation may work well enough for its immediate task while the composite route changes a validity-relevant predicate. The resulting conflict should not be diagnosed automatically as irrationality, deception, or logical contradiction. First test the translation path, the erased indices, and the comparator by which the conflict was declared.

EMPIRICAL / SOCIAL BRIDGE

The claim that large communicative networks actually produce psychological or institutional “ripples” through noncommuting standards is empirical. K21 only proves that noncommutation is possible and therefore must not be ruled out by an assumption that all successful local translations compose into one globally coherent frame.

19. Machine verification as a calibrated participant

The semantic-deflation companion provides a concrete example of the distinction this paper is trying to preserve. It formalizes two predicates under one public label and proves both the scoped branch statements and the failure of meaning preservation. Its Lean Web receipt reports that the declarations typechecked in the captured environment [5]. The formal system did not confuse the predicates. The public label was the lossy projection.

K18-K19 reproduce a deliberately smaller version inside the present source. Both Usage.original and Usage.extended map to the string “singularity,” while tinyMeaning gives them different values on a marked witness. The formal lesson is not that the word singularity is defective. It is that equality of public labels does not entail equality of the predicates or semantic contracts supplying their use.

LEAN CALIBRATION - POSITIVE, NOT ADVERSARIAL

This paper does not present semantic boundaries as shortcomings of Lean. A proof assistant becomes more valuable when its jurisdiction is clear. By forcing definitions, hypotheses, and type distinctions into inspectable form, Lean can expose exactly where an argument depends on a bridge that remains outside the encoded theorem. The verifier is strongest when a user does not ask a proof receipt to certify an unsubmitted semantic claim.

The same discipline applies in the other direction. A linguistic or philosophical question should not be dismissed merely because it has not yet been encoded. The correct status may be OPEN, bridge-pending, empirically testable, ill-typed under the current registry, or simply outside the current proof task. Formalization is a powerful transformation, not a magical transfer of every meaning in the source sentence into kernel syntax.

20. Human-machine cooperative descriptive intelligence

The constructive consequence is not to oppose human and machine as rival claimants to one privileged intelligence. The language paper already treats contemporary language models as contrast instruments: they can generate alternate representations quickly enough to make hidden assumptions, semantic drift, and mode changes inspectable, while their outputs still inherit training distributions, prompts, system constraints, and architecture [1]. That is a useful capability precisely because it is not a view from nowhere.

A cooperative architecture can divide labor without assigning moral polarity to the self/other distinction. Human participants can contribute lived stakes, goals, value judgments, local context, embodied consequence, and residual selections. Machine systems can preserve large dependency graphs, compare alternate formulations, track definitions, search for counterexamples, produce formal translations, maintain receipts, and detect noncommuting bridges. Either side can also fail. A design that exposes provenance and disagreement makes those failures easier to localize.

DESIGN DIRECTION - RELATIONAL SUPERINTELLIGENCE

Use “superintelligence” here, if at all, for a cooperative capacity emerging from linked human and machine processes whose strengths remain typed and mutually inspectable. Do not pre-classify the technological other as an enemy because it is other, or as a benevolent oracle that should privilege the current self. Otherness supplies contrast; antagonism, trust, authority, and care require additional predicates and evidence.

This is also where the pharmakon can be turned recursively upon itself. Language compresses frames and can hide them; language technology can generate alternate framings, explicit bridge ledgers, source hashes, proof objects, countermodels, and translation comparisons that make some of that hidden machinery visible again. The recursion is not escaped. It is instrumented.

21. Attack surface and falsification conditions

The framework should fail locally where its declared premises fail. The following surfaces are intentionally exposed rather than protected by rhetorical scope changes.

- Reference bridge: exhibit a form of meaningful reference for which source-role/target-role discrimination is neither present nor replaceable by any functionally equivalent distinction, while the same referential content remains recoverable.
- Self/other terminology: show that identifying source-side with self-side and target-side with other-side produces contradictions or loses a distinction essential to the intended application. That would defeat the bridge vocabulary without refuting K01-K04.
- Criterion/exclusion: give an exclusion claim in the admitted class whose excluded status requires no criterion, complement, boundary, or inferential rule. Alternatively show that the paper has illicitly generalized a local complement construction to a use of “metaphysical” outside that class.
- Lambda fibre result: refute K08 or K09 by identifying a formal error, or show that a purported hidden distinction is actually target-irrelevant so the factorization obstruction does not apply.
- Residual will: derive a unique terminal selector from the registered predecessor conditions in a case that the paper classifies as residual. The residue then disappears at that scope. This is a correction, not a threat to the method.

- No-first bridge: provide a coherent space-first claim in the admitted class that establishes priority without the temporal-type resource the paper attributes to it, or a time-first claim that establishes priority without individuation/localization. K15 would remain valid conditionally while the physical bridge application would fail.
- Absolute-jurisdiction argument: supply an explicit invariance/correspondence theorem that genuinely carries a local result across the quantified frame class. That would extend warrant rather than refute local validity.
- Attention bridge: empirical evidence may show that actual cognition does not allocate resources in the proposed immersive way, even though K22 remains elementary arithmetic.
- Communication model: demonstrate that the proposed predicate-preservation handshake misses the operative semantic relation, or that a claimed network conflict occurs despite a commuting and meaning-preserving translation diagram.
- Human-machine design direction: evaluate it by actual performance, safety, error detection, collaboration quality, and preservation of dissent. It should not receive theorem status by association with the formal kernel.

22. Claims, non-claims, and stopping condition

The paper stops when the formal condensates have explicit translations, the bridge obligations have been separated, and the constructive consequences can be stated without promotion beyond their warrant.

22.1 Claimed

- The 23 formal condensates are mapped to exact Lean declarations in the attached source and translation ledger.
- Reference polarity, fibre obstruction, residual selection, conditional no-first-position, local/universal warrant separation, translation preservation, noncommutation possibility, resource tradeoff, and interpretation variance follow under the declared formal definitions and premises. Appendix B records acceptance of their Lean declarations.
- The No-First-Position result is conditional on explicit spatiotemporal bridge premises; the paper argues for those premises in the specified class rather than disguising them as kernel facts.
- Compression can remain exact for a chosen target while losing other distinctions, so successful closure for one task does not imply source exhaustion.
- The proper use of machine verification requires preserving the distinction between the encoded theorem and the semantic or physical bridge by which it is applied.

22.2 Not claimed

- That Lean or any theorem prover is defective because it checks formal propositions rather than unencoded meanings.
- That physical space and time literally generate one another, are identical, or have no physical ontology.
- That self/other is proven to be the observer-independent first distinction of reality.
- That residual will proves consciousness, libertarian free will, randomness, or absence of mechanism.

- That “metaphysical” has one universal lexical definition; only the exclusionary nonphysical use is audited through complement structure.
- That frame-indexed warrant implies relativism, anti-realism, or weak evidence.
- That human attention, cultural conflict, or human-machine cooperation follows from logic alone.
- That the captured execution receipt, by itself, certifies an unencoded linguistic, physical, cultural, or ontological interpretation.

23. Conclusion: bounded warrant does not mean weak warrant

The argument ends where it began: a description earns precision through constraints. A predicate gains discriminatory power by excluding alternatives. A coordinate becomes useful by fixing relations. A theorem becomes checkable by declaring syntax, types, definitions, and rules. A measurement becomes reproducible by fixing a comparator, calibration, and tolerance. Those boundaries do not make the result worthless. They are the conditions through which the result can become exact.

The error begins when a successful local closure is asked to erase the very frame that gave it operational meaning. A claim about “the absolute” can be syntactically and semantically meaningful inside a shared logical frame; what it cannot do merely by naming its target is make the act of assertion self-certifying across every possible frame. A rigorous result can therefore be both deeply valuable and explicitly scoped. Bounded warrant does not mean weak warrant.

Reference shows the same recursive shape. A prior source can become a successor referent, but the successor act again distinguishes source-role from target-role. A criterion can be selected, a complement formed, an “outside” named, and the previous boundary brought inside a larger atlas. The recursion does not prove that some final object lies forever beyond description. It shows that objectifying a prior framing condition does not, by that operation alone, eliminate the current act of framing.

Will occupies the residual seam without being inflated into a metaphysical agent. Where the predecessor registry does not uniquely force one admissible continuation or criterion, the selected distinction remains additional to that predecessor account. Later mechanism may absorb the residue. Until then, the honest status is neither miracle nor deterministic fiction; it is an unattributed selection relative to the declared registry.

Language then appears as both medicine and poison in a precise technological sense. It compresses immense histories of operation into portable primitives, shared customs, logical handshakes, and public labels. This enables coordination and conserves finite attention. It can also hide the bridges through which those primitives acquired their meaning. The response is not to abolish compression, standards, proof, or machine verification. It is to make important closures reopenable, preserve their indices and receipts, and teach our tools to expose alternative framings without pretending that any tool stands outside framing.

FINAL CONDENSATE

Reference requires a maintained polarity in the model that carries it. Criteria make partitions and exclusions operational. Residual will name selection not already forced by the registered predecessor conditions. Projection can be exact for a target without sequestering what it omits. Under declared spatiotemporal bridge premises, neither “space first” nor “time first” earns complement-free absolute descriptive priority. Formal verification can certify these encoded relations with extraordinary precision when the translation boundary remains visible. That visibility does not diminish rigor; it is part of the rigor.

The constructive opportunity follows directly. Better communication need not demand one universal tongue or one permanently privileged frame. It can use explicit semantic handshakes, preserved provenance, adversarial contrast, formal receipts, and translation tests to discover where apparently shared words stop carrying shared structure. Human and machine capabilities can then cooperate at that boundary. The resulting intelligence need not turn otherness into hostility or selfhood into privilege. It can treat distinction as the condition for comparison and comparison as an opportunity for correction.

23.1 Operational protocol for corrigible communication

The theory closes constructively by turning its own boundary discipline into a communication protocol. The protocol is deliberately compatible with ordinary compressed language: it is invoked when stakes, disagreement, novelty, or translation cost justify reopening the frame.

- Name the current referent and, where relevant, the source-role from which it is being referred to.
- State the criterion that creates the operative distinction and identify the comparator that will adjudicate it.
- Preserve frame indices - domain, time, scale, mode, interpretation, and jurisdiction - whenever erasing them could change validity.
- Distinguish a formal derivation from the bridge that maps its symbols to an informal or physical target.
- When a conclusion is compressed into a status word such as true, false, verified, objective, safe, dangerous, self, other, or absolute, retain a route back to the conditions that licensed the label.
- When two communities or systems disagree, test whether their translations preserve the same predicates and whether alternative translation paths commute before treating the disagreement as one proposition plus its negation.
- Where the predecessor registry does not uniquely determine the next criterion, threshold, verifier, or action, mark the residual selection rather than silently attributing it to mechanism, randomness, authority, or personhood.
- Use human and machine contrast iteratively: propose alternate formulations, formalize what can be formalized, verify the encoded kernel, inspect residue, and reopen only the boundaries material to the next decision.

STOPPING RULE FOR PRACTICE

Communication need not reproduce every hidden dependency before any action can occur. Reopen a compressed frame when a validity-relevant distinction, disagreement, safety consequence, translation failure, or new target makes the omitted structure material. Otherwise local closure can remain operative and revisable.

Appendix A. Complete accepted Lean source

The source below contains the accepted declarations. K06 includes the explicit witness type X required by Lean. The embedded `LambdaSpatiotemporalFrame.lean` preserves the exact accepted UTF-8 bytes. Appendix B records execution and reproduction.

```
import Lean

/-!
# Lambda, Reference, and the Spatiotemporal Descriptive Frame
Formal kernel for the companion paper.

Scope:
This file checks only the propositions encoded below. The terms `self`, `other`,
`space`, `time`, `will`, `absolute`, and `meaning` receive only the formal roles
explicitly assigned here. Physical, phenomenological, linguistic, or ontological
interpretations require separate bridge arguments in the paper.
-/

namespace LambdaSpatiotemporal

/-! ## 1. Reference polarity -/

inductive RefRole where
| source
| target
deriving DecidableEq, Repr

structure RoleTagged (X : Type) where
carrier : X
role : RefRole
deriving DecidableEq, Repr

def asSource {X : Type} (x : X) : RoleTagged X := {x, .source}
def asTarget {X : Type} (x : X) : RoleTagged X := {x, .target}

theorem source_target_roles_distinct : RefRole.source ≠ RefRole.target := by
intro h
cases h

theorem reference_polarity {X : Type} (x y : X) :
asSource x ≠ asTarget y := by
intro h
have hr : RefRole.source = RefRole.target := congrArg RoleTagged.role h
exact source_target_roles_distinct hr

/-- Even when the carrier is the same, referring and referred-to roles remain distinct. -/
theorem reflexive_reference_preserves_role_distinction {X : Type} (x : X) :
asSource x ≠ asTarget x :=
reference_polarity x x

/-- Repeating the tagging operation at any indexed station preserves role polarity. -/
theorem iterated_reference_polarity {X : Type} (x : X) (n : Nat) :
(n, asSource x) ≠ (n, asTarget x) := by
intro h
have ht : asSource x = asTarget x := congrArg Prod.snd h
exact reflexive_reference_preserves_role_distinction x ht

/-! ## 2. Binary partition and self/other bridge -/

def SelfSide {X : Type} (criterion : X → Bool) (x : X) : Prop := criterion x = true
def OtherSide {X : Type} (criterion : X → Bool) (x : X) : Prop := criterion x = false
```

Appendix A. Accepted Lean source (continued)

```

def NontrivialCriterion {X : Type} (criterion : X → Bool) : Prop :=
  (∃ x, SelfSide criterion x) ∧ (∃ y, OtherSide criterion y)

theorem nontrivial_criterion_exhibits_poles {X : Type} (criterion : X → Bool)
  (h : NontrivialCriterion criterion) :
  ∃ x y, SelfSide criterion x ∧ OtherSide criterion y := by
  rcases h with ⟨(x, hx), (y, hy)⟩
  exact ⟨x, y, hx, hy⟩

theorem nontrivial_criterion_exhibits_distinct_states {X : Type} (criterion : X → Bool)
  (h : NontrivialCriterion criterion) :
  ∃ x y : X, x ≠ y := by
  rcases h with ⟨(x, hx), (y, hy)⟩
  refine ⟨x, y, ?_⟩
  intro hxy
  subst y
  have : true = false := hx.symm.trans hy
  cases this

/-! ## 3. Exclusion is indexed to a criterion -/

def ExcludedBy {X : Type} (criterion : X → Prop) (x : X) : Prop := ¬ criterion x

theorem exclusion_is_criterion_relative {X : Type} (criterion : X → Prop) (x : X) :
  ExcludedBy criterion x ↔ ¬ criterion x :=
  Iff.rfl

/-! ## 4. Lambda projection / fibre obstruction -/

theorem postprocessing_cannot_split_fibre
  {X Y Z : Type} (projection : X → Y) (g : Y → Z) {x₁ x₂ : X}
  (h : projection x₁ = projection x₂) :
  g (projection x₁) = g (projection x₂) :=
  congrArg g h

def FactorsThrough {X Y Z : Type} (projection : X → Y) (target : X → Z) : Prop :=
  ∃ g : Y → Z, ∀ x, g (projection x) = target x

theorem target_difference_blocks_factorization
  {X Y Z : Type} (projection : X → Y) (target : X → Z)
  {x₁ x₂ : X} (hproj : projection x₁ = projection x₂)
  (htarget : target x₁ ≠ target x₂) :
  ¬ FactorsThrough projection target := by
  intro hfac
  rcases hfac with ⟨g, hg⟩
  apply htarget
  calc
    target x₁ = g (projection x₁) := (hg x₁).symm
    _ = g (projection x₂) := congrArg g hproj
    _ = target x₂ := hg x₂

/-! ## 5. Compression may be exact for a target while remaining non-injective -/

def InjectiveMap {X Y : Type} (f : X → Y) : Prop :=
  ∀ ⟨x y⟩, f x = f y → x = y

theorem target_exact_compression_can_lose_other_distinctions :
  ∃ (compress : Bool × Bool → Bool) (target : Bool × Bool → Bool),
  FactorsThrough compress target ∧ ¬ InjectiveMap compress := by
  let compress : Bool × Bool → Bool := fun p => p.1
  let target : Bool × Bool → Bool := fun p => p.1
  refine ⟨compress, target, ?_, ?_⟩

```

Appendix A. Accepted Lean source (continued)

```

· refine ((fun b => b), ?_)
  intro x
  rfl
· intro hinj
  have hpair : (false, false) = (false, true) := hinj rfl
  have hsnd : false = true := congrArg Prod.snd hpair
  cases hsnd

/--! ## 6. Residual selection / will kernel -/

def UniqueOutcome {J S : Type} (Admissible : J → Prop) (outcome : J → S) : Prop :=
  ∃ s : S, ∀ j, Admissible j → outcome j = s

theorem residual_selection
  {J S : Type} (Admissible : J → Prop) (outcome : J → S)
  {j₁ j₂ : J} (h₁ : Admissible j₁) (h₂ : Admissible j₂)
  (hdiff : outcome j₁ ≠ outcome j₂) :
  ¬ UniqueOutcome Admissible outcome := by
  intro huniq
  rcases huniq with (s, hs)
  apply hdiff
  calc
    outcome j₁ = s := hs j₁ h₁
    _ = outcome j₂ := (hs j₂ h₂).symm

/-- Formal role used by the paper's term `will`: an admissible selected procedure
    in a registry whose admissible procedures do not determine one unique outcome. -/
def ResidualWill {J S : Type} (Admissible : J → Prop) (outcome : J → S) (chosen : J) : Prop :=
  Admissible chosen ∧ ¬ UniqueOutcome Admissible outcome

theorem residual_will_from_competing_admissible_procedures
  {J S : Type} (Admissible : J → Prop) (outcome : J → S)
  {chosen j₂ : J} (hchosen : Admissible chosen) (h₂ : Admissible j₂)
  (hdiff : outcome chosen ≠ outcome j₂) :
  ResidualWill Admissible outcome chosen := by
  exact (hchosen, residual_selection Admissible outcome hchosen h₂ hdiff)

/-- A selected procedure can index a criterion without the criterion being uniquely forced by
    the predecessor registry. The substantive premise is that two admissible procedures induce
    different criteria. -/
def SelectedCriterion {J X : Type} (criterionOf : J → X → Bool) (chosen : J) : X → Bool :=
  criterionOf chosen

theorem competing_admissible_criteria_not_uniquely_forced
  {J X : Type} (Admissible : J → Prop) (criterionOf : J → X → Bool)
  {j₁ j₂ : J} (h₁ : Admissible j₁) (h₂ : Admissible j₂)
  (hdiff : criterionOf j₁ ≠ criterionOf j₂) :
  ¬ UniqueOutcome Admissible criterionOf :=
  residual_selection Admissible criterionOf h₁ h₂ hdiff

/--! ## 7. Absolute-priority schema and no-first-position kernel -/

def AbsolutePriority {C : Type} (Priority UsesComplement : C → Prop) (c : C) : Prop :=
  Priority c ∧ ¬ UsesComplement c

theorem complement_requirement_blocks_absolute_priority
  {C : Type} (Priority UsesComplement : C → Prop)
  (hbridge : ∀ c, Priority c → UsesComplement c) :
  ¬ ∃ c, AbsolutePriority Priority UsesComplement c := by
  intro h
  rcases h with (c, hp, hnot)
  exact hnot (hbridge c hp)

```

Appendix A. Accepted Lean source (continued)

```

structure SpatiotemporalBridge (Claim : Type) where
  SpaceFirst : Claim → Prop
  TimeFirst : Claim → Prop
  UsesTemporalResource : Claim → Prop
  UsesSpatialResource : Claim → Prop
  spaceFirstUsesTemporal : ∀ c, SpaceFirst c → UsesTemporalResource c
  timeFirstUsesSpatial : ∀ c, TimeFirst c → UsesSpatialResource c

def AbsoluteSpaceFirst {Claim : Type} (B : SpatiotemporalBridge Claim) (c : Claim) : Prop :=
  AbsolutePriority B.SpaceFirst B.UsesTemporalResource c

def AbsoluteTimeFirst {Claim : Type} (B : SpatiotemporalBridge Claim) (c : Claim) : Prop :=
  AbsolutePriority B.TimeFirst B.UsesSpatialResource c

theorem no_first_position {Claim : Type} (B : SpatiotemporalBridge Claim) :
  (¬ ∃ c, AbsoluteSpaceFirst B c) ∧ (¬ ∃ c, AbsoluteTimeFirst B c) := by
  constructor
  · exact complement_requirement_blocks_absolute_priority
    B.SpaceFirst B.UsesTemporalResource B.spaceFirstUsesTemporal
  · exact complement_requirement_blocks_absolute_priority
    B.TimeFirst B.UsesSpatialResource B.timeFirstUsesSpatial

/-! ## 8. Frame-local warrant does not self-promote to universal warrant -/

inductive TwoFrame where
| local
| other
deriving DecidableEq, Repr

def SampleWarrant : TwoFrame → Prop
| .local => True
| .other => False

def UniversalWarrant {F : Type} (Warrant : F → Prop) : Prop := ∀ f, Warrant f

theorem local_warrant_not_universal_in_general :
  SampleWarrant .local ∧ ¬ UniversalWarrant SampleWarrant := by
  constructor
  · exact True.intro
  · intro hall
    exact hall .other

/-- A predicate named or intended as `absolute` has no elimination rule by logic alone. -/
theorem absolute_label_does_not_self_license :
  ¬ (∀ (Absolute : Prop → Prop) (p : Prop), Absolute p → p) := by
  intro h
  have hfalse : False := h (fun _ => True) False True.intro
  exact hfalse

/-! ## 9. Public-label compression can suppress operative meaning -/

inductive Usage where
| original
| extended
deriving DecidableEq, Repr

inductive TinyState where
| bare
| marked
deriving DecidableEq, Repr

```

Appendix A. Accepted Lean source (continued)

```

def tinyMeaning : Usage → TinyState → Bool
| .original, .bare => true
| .original, .marked => false
| .extended, _ => true

def publicLabel (_ : Usage) : String := "singularity"

theorem same_public_label : publicLabel .original = publicLabel .extended := rfl

theorem same_label_can_hide_different_meanings :
  publicLabel .original = publicLabel .extended ∧
  tinyMeaning .original .marked ≠ tinyMeaning .extended .marked := by
  constructor
  · rfl
  · decide

/-! ## 10. Translation bridges: preservation and noncommutation -/

def PreservesPredicate {A B : Type} (f : A → B) (P : A → Prop) (Q : B → Prop) : Prop :=
  ∀ a, P a ↔ Q (f a)

theorem predicate_preservation_composes
  {A B C : Type} (f : A → B) (g : B → C)
  (P : A → Prop) (Q : B → Prop) (R : C → Prop)
  (hAB : PreservesPredicate f P Q)
  (hBC : PreservesPredicate g Q R) :
  PreservesPredicate (fun a => g (f a)) P R := by
  intro a
  exact Iff.trans (hAB a) (hBC (f a))

theorem translations_need_not_commute :
  ∃ (AB BC AC : Bool → Bool) (x : Bool), AC x ≠ BC (AB x) := by
  refine ⟨(fun b => b), (fun b => b), Bool.not, false, ?_⟩
  decide

/-! ## 11. Finite resource budget forces a tradeoff when full demand exceeds it -/

theorem finite_budget_forces_tradeoff
  (costA costB allocA allocB budget : Nat)
  (hDemand : budget < costA + costB)
  (hBudget : allocA + allocB ≤ budget)
  (hA : costA ≤ allocA)
  (hB : costB ≤ allocB) : False := by
  have hsum : costA + costB ≤ allocA + allocB := Nat.add_le_add hA hB
  have hfit : costA + costB ≤ budget := Nat.le_trans hsum hBudget
  exact (Nat.not_le_of_lt hDemand) hfit

/-! ## 12. Frozen inscription does not by logic alone freeze interpretation -/

theorem same_input_can_receive_different_interpretations :
  ∃ (I₀ I₁ : Bool → Bool) (r : Bool), I₀ r ≠ I₁ r := by
  refine ⟨(fun b => b), Bool.not, false, ?_⟩
  decide

/-! ## 13. Audit checks -/

#check source_target_roles_distinct
#check reference_polarity
#check reflexive_reference_preserves_role_distinction
#check iterated_reference_polarity
#check nontrivial_criterion_exhibits_poles
#check nontrivial_criterion_exhibits_distinct_states

```

Appendix A. Accepted Lean source (continued)

```

#check exclusion_is_criterion_relative
#check postprocessing_cannot_split_fibre
#check target_difference_blocks_factorization
#check target_exact_compression_can_lose_other_distinctions
#check residual_selection
#check residual_will_from_competing_admissible_procedures
#check SelectedCriterion
#check competing_admissible_criteria_not_uniquely_forced
#check complement_requirement_blocks_absolute_priority
#check no_first_position
#check local_warrant_not_universal_in_general
#check absolute_label_does_not_self_license
#check same_public_label
#check same_label_can_hide_different_meanings
#check predicate_preservation_composes
#check translations_need_not_commute
#check finite_budget_forces_tradeoff
#check same_input_can_receive_different_interpretations

#print axioms source_target_roles_distinct
#print axioms reference_polarity
#print axioms reflexive_reference_preserves_role_distinction
#print axioms iterated_reference_polarity
#print axioms nontrivial_criterion_exhibits_poles
#print axioms nontrivial_criterion_exhibits_distinct_states
#print axioms exclusion_is_criterion_relative
#print axioms postprocessing_cannot_split_fibre
#print axioms target_difference_blocks_factorization
#print axioms target_exact_compression_can_lose_other_distinctions
#print axioms residual_selection
#print axioms residual_will_from_competing_admissible_procedures
#print axioms competing_admissible_criteria_not_uniquely_forced
#print axioms complement_requirement_blocks_absolute_priority
#print axioms no_first_position
#print axioms local_warrant_not_universal_in_general
#print axioms absolute_label_does_not_self_license
#print axioms same_public_label
#print axioms same_label_can_hide_different_meanings
#print axioms predicate_preservation_composes
#print axioms translations_need_not_commute
#print axioms finite_budget_forces_tradeoff
#print axioms same_input_can_receive_different_interpretations

#eval Lean.versionString
#eval "LAMBDA_SPATIOTEMPORAL_TRANSLATION_AUDIT_COMPLETE"

end LambdaSpatiotemporal

```

Appendix B. Lean execution receipts

PASS - 23 theorem declarations accepted

Executed environment	Lean Web / mathlib-stable / Lean 4.33.0
Selected release	Stable Release (v4.33.0 with mathlib, cslib)
Source import	Lean only
Captured at	10 September 2026, 22:48:56 UTC
Declaration checks	23 theorems + SelectedCriterion = 24
Axiom reports	23/23: no axiom dependencies
Final diagnostics	0 errors; 0 warnings
Message accounting	24 checks + 23 axiom reports + 2 evaluations = 49
Accepted source	13,340 UTF-8 bytes; 352 lines; exact editor match

Completion marker

LAMBDA_SPATIOTEMPORAL_TRANSLATION_AUDIT_COMPLETE

Correction recorded before acceptance

The original K06 header omitted the domain of its existential witnesses. Lean emitted six diagnostics and reported sorryAx for that failed declaration. That run supplies diagnostic history, not acceptance. The only source edit adds : X at line 69:

Before: $\exists x y, x \neq y := by$

After: $\exists x y : X, x \neq y := by$

The full corrected source produced the final 49-message receipt. Each final axiom report states that its theorem depends on no axioms. This result retains every explicit hypothesis in the declarations, including the SpatiotemporalBridge premises. The theorem statements and their bridges keep the scope declared in Section 3.

Receipt basis: observed Lean Web output and exact source comparison. This record does not assert a local compiler process exit code or a separately signed certification.

Appendix B. Declaration-by-declaration receipt

The ledger IDs match Section 3. Every listed theorem has an accepted #check output and an axiom report with no dependencies. The namespace is LambdaSpatiotemporal.

ID	DECLARATION	RESULT
K01	source_target_roles_distinct	PASS
K02	reference_polarity	PASS
K03	reflexive_reference_preserves_role_distinction	PASS
K04	iterated_reference_polarity	PASS
K05	nontrivial_criterion_exhibits_poles	PASS
K06	nontrivial_criterion_exhibits_distinct_states	PASS
K07	exclusion_is_criterion_relative	PASS
K08	postprocessing_cannot_split_fibre	PASS
K09	target_difference_blocks_factorization	PASS
K10	target_exact_compression_can_lose_other_distinctions	PASS
K11	residual_selection	PASS
K12	residual_will_from_competing_admissible_procedures	PASS
K13	competing_admissible_criteria_not_uniquely_forced	PASS
K14	complement_requirement_blocks_absolute_priority	PASS
K15	no_first_position	PASS
K16	local_warrant_not_universal_in_general	PASS
K17	absolute_label_does_not_self_license	PASS
K18	same_public_label	PASS
K19	same_label_can_hide_different_meanings	PASS
K20	predicate_preservation_composes	PASS
K21	translations_need_not_commute	PASS
K22	finite_budget_forces_tradeoff	PASS
K23	same_input_can_receive_different_interpretations	PASS

Additional definition check: SelectedCriterion - PASS. The next pages reproduce the entire final All Messages text; automatic line wrapping affects layout only. The embedded text attachment preserves the captured transcript.

Appendix B. Complete final Lean Web transcript

```

mathlib-stable.lean:353:0
No info found.
All Messages (49)
mathlib-stable.lean:300:0
LambdaSpatiotemporal.source_target_roles_distinct : RefRole.source ≠ RefRole.target
mathlib-stable.lean:301:0
LambdaSpatiotemporal.reference_polarity {X : Type} (x y : X) : asSource x ≠ asTarget y
mathlib-stable.lean:302:0
LambdaSpatiotemporal.reflexive_reference_preserves_role_distinction {X : Type} (x : X) : asSource x ≠
asTarget x
mathlib-stable.lean:303:0
LambdaSpatiotemporal.iterated_reference_polarity {X : Type} (x : X) (n : Nat) : (n, asSource x) ≠ (n,
asTarget x)
mathlib-stable.lean:304:0
LambdaSpatiotemporal.nontrivial_criterion_exhibits_poles {X : Type} (criterion : X → Bool)
(h : NontrivialCriterion criterion) : ∃ x y, SelfSide criterion x ∧ OtherSide criterion y
mathlib-stable.lean:305:0
LambdaSpatiotemporal.nontrivial_criterion_exhibits_distinct_states {X : Type} (criterion : X → Bool)
(h : NontrivialCriterion criterion) : ∃ x y, x ≠ y
mathlib-stable.lean:306:0
LambdaSpatiotemporal.exclusion_is_criterion_relative {X : Type} (criterion : X → Prop) (x : X) :
ExcludedBy criterion x ↔ ¬criterion x
mathlib-stable.lean:307:0
LambdaSpatiotemporal.postprocessing_cannot_split_fibre {X Y Z : Type} (projection : X → Y) (g : Y → Z)
{x₁ x₂ : X}
(h : projection x₁ = projection x₂) : g (projection x₁) = g (projection x₂)
mathlib-stable.lean:308:0
LambdaSpatiotemporal.target_difference_blocks_factorization {X Y Z : Type} (projection : X → Y) (target
: X → Z)
{x₁ x₂ : X} (hproj : projection x₁ = projection x₂) (htarget : target x₁ ≠ target x₂) :
¬FactorsThrough projection target
mathlib-stable.lean:309:0
LambdaSpatiotemporal.target_exact_compression_can Lose_other_distinctions :
∃ compress target, FactorsThrough compress target ∧ ¬InjectiveMap compress
mathlib-stable.lean:310:0
LambdaSpatiotemporal.residual_selection {J S : Type} (Admissible : J → Prop) (outcome : J → S) {j₁ j₂ :
J}
(h₁ : Admissible j₁) (h₂ : Admissible j₂) (hdiff : outcome j₁ ≠ outcome j₂) : ¬UniqueOutcome
Admissible outcome
mathlib-stable.lean:311:0
LambdaSpatiotemporal.residual_will_from_competing_admissible_procedures {J S : Type} (Admissible : J →
Prop)
(outcome : J → S) {chosen j₂ : J} (hchosen : Admissible chosen) (h₂ : Admissible j₂)
(hdiff : outcome chosen ≠ outcome j₂) : ResidualWill Admissible outcome chosen
mathlib-stable.lean:312:0
LambdaSpatiotemporal.SelectedCriterion {J X : Type} (criterionOf : J → X → Bool) (chosen : J) : X →
Bool
mathlib-stable.lean:313:0
LambdaSpatiotemporal.competing_admissible_criteria_not_uniquely_forced {J X : Type} (Admissible : J →
Prop)
(criterionOf : J → X → Bool) {j₁ j₂ : J} (h₁ : Admissible j₁) (h₂ : Admissible j₂)
(hdiff : criterionOf j₁ ≠ criterionOf j₂) : ¬UniqueOutcome Admissible criterionOf
mathlib-stable.lean:314:0
LambdaSpatiotemporal.complement_requirement_blocks_absolute_priority {C : Type} (Priority
UsesComplement : C → Prop)
(hbridge : ∀ (c : C), Priority c → UsesComplement c) : ¬∃ c, AbsolutePriority Priority UsesComplement
c
mathlib-stable.lean:315:0
LambdaSpatiotemporal.no_first_position {Claim : Type} (B : SpatiotemporalBridge Claim) :
(¬∃ c, AbsoluteSpaceFirst B c) ∧ ¬∃ c, AbsoluteTimeFirst B c
mathlib-stable.lean:316:0
LambdaSpatiotemporal.local_warrant_not_universal_in_general :
SampleWarrant TwoFrame.local ∧ ¬UniversalWarrant SampleWarrant

```

Appendix B. Final transcript (continued)

```

mathlib-stable.lean:317:0
LambdaSpatiotemporal.absolute_label_does_not_self_license :  $\neg \forall$  (Absolute : Prop  $\rightarrow$  Prop) (p : Prop),
Absolute p  $\rightarrow$  p
mathlib-stable.lean:318:0
LambdaSpatiotemporal.same_public_label : publicLabel Usage.original = publicLabel Usage.extended
mathlib-stable.lean:319:0
LambdaSpatiotemporal.same_label_can_hide_different_meanings :
  publicLabel Usage.original = publicLabel Usage.extended  $\wedge$ 
    tinyMeaning Usage.original TinyState.marked  $\neq$  tinyMeaning Usage.extended TinyState.marked
mathlib-stable.lean:320:0
LambdaSpatiotemporal.predicate_preservation_composes {A B C : Type} (f : A  $\rightarrow$  B) (g : B  $\rightarrow$  C) (P : A  $\rightarrow$ 
Prop)
  (Q : B  $\rightarrow$  Prop) (R : C  $\rightarrow$  Prop) (hAB : PreservesPredicate f P Q) (hBC : PreservesPredicate g Q R) :
  PreservesPredicate (fun a  $\Rightarrow$  g (f a)) P R
mathlib-stable.lean:321:0
LambdaSpatiotemporal.translations_need_not_commute :  $\exists$  AB BC AC x, AC x  $\neq$  BC (AB x)
mathlib-stable.lean:322:0
LambdaSpatiotemporal.finite_budget_forces_tradeoff (costA costB allocA allocB budget : Nat)
  (hDemand : budget < costA + costB) (hBudget : allocA + allocB  $\leq$  budget) (hA : costA  $\leq$  allocA) (hB :
costB  $\leq$  allocB) :
  False
mathlib-stable.lean:323:0
LambdaSpatiotemporal.same_input_can_receive_different_interpretations :  $\exists$  I0 I1 r, I0 r  $\neq$  I1 r
mathlib-stable.lean:325:0
'LambdaSpatiotemporal.source_target_roles_distinct' does not depend on any axioms
mathlib-stable.lean:326:0
'LambdaSpatiotemporal.reference_polarity' does not depend on any axioms
mathlib-stable.lean:327:0
'LambdaSpatiotemporal.reflexive_reference_preserves_role_distinction' does not depend on any axioms
mathlib-stable.lean:328:0
'LambdaSpatiotemporal.iterated_reference_polarity' does not depend on any axioms
mathlib-stable.lean:329:0
'LambdaSpatiotemporal.nontrivial_criterion_exhibits_poles' does not depend on any axioms
mathlib-stable.lean:330:0
'LambdaSpatiotemporal.nontrivial_criterion_exhibits_distinct_states' does not depend on any axioms
mathlib-stable.lean:331:0
'LambdaSpatiotemporal.exclusion_is_criterion_relative' does not depend on any axioms
mathlib-stable.lean:332:0
'LambdaSpatiotemporal.postprocessing_cannot_split_fibre' does not depend on any axioms
mathlib-stable.lean:333:0
'LambdaSpatiotemporal.target_difference_blocks_factorization' does not depend on any axioms
mathlib-stable.lean:334:0
'LambdaSpatiotemporal.target_exact_compression_can_lose_other_distinctions' does not depend on any
axioms
mathlib-stable.lean:335:0
'LambdaSpatiotemporal.residual_selection' does not depend on any axioms
mathlib-stable.lean:336:0
'LambdaSpatiotemporal.residual_will_from_competing_admissible_procedures' does not depend on any axioms
mathlib-stable.lean:337:0
'LambdaSpatiotemporal.competing_admissible_criteria_not_uniquely_forced' does not depend on any axioms
mathlib-stable.lean:338:0
'LambdaSpatiotemporal.complement_requirement_blocks_absolute_priority' does not depend on any axioms
mathlib-stable.lean:339:0
'LambdaSpatiotemporal.no_first_position' does not depend on any axioms
mathlib-stable.lean:340:0
'LambdaSpatiotemporal.local_warrant_not_universal_in_general' does not depend on any axioms
mathlib-stable.lean:341:0
'LambdaSpatiotemporal.absolute_label_does_not_self_license' does not depend on any axioms
mathlib-stable.lean:342:0
'LambdaSpatiotemporal.same_public_label' does not depend on any axioms
mathlib-stable.lean:343:0
'LambdaSpatiotemporal.same_label_can_hide_different_meanings' does not depend on any axioms
mathlib-stable.lean:344:0

```

Appendix B. Final transcript (continued)

```
'LambdaSpatiotemporal.predicate_preservation_composes' does not depend on any axioms
mathlib-stable.lean:345:0
'LambdaSpatiotemporal.translations_need_not_commute' does not depend on any axioms
mathlib-stable.lean:346:0
'LambdaSpatiotemporal.finite_budget_forces_tradeoff' does not depend on any axioms
mathlib-stable.lean:347:0
'LambdaSpatiotemporal.same_input_can_receive_different_interpretations' does not depend on any axioms
mathlib-stable.lean:349:0
"4.33.0"
mathlib-stable.lean:350:0
"LAMBDA_SPATIOTEMPORAL_TRANSLATION_AUDIT_COMPLETE"
Restart File
```

Appendix B. Reproduction and source identity

Reproduce the accepted run

1. Open the embedded `REPRODUCE_LEAN_WEB.url`, or visit [Lean Web](#) and select Stable Release (v4.33.0 with `mathlib`, `cslib`), project `mathlib-stable`.
2. Load the complete embedded `LambdaSpatiotemporalFrame.lean` into the editor. Wait for processing to finish, then expand All Messages.
3. Require all 24 declaration checks, all 23 axiom reports, the version string 4.33.0, and the completion marker. Require zero errors and warnings. A marker alone does not establish acceptance.
4. Compare the complete editor source with the embedded file. This edition compared the full editor selection with the frozen source and obtained an exact match.
5. For any later source edit, execute the changed file again and capture a new receipt. A PDF layout or prose edit around an unchanged source does not alter the Lean declarations.

Embedded evidence

`LambdaSpatiotemporalFrame.lean`
Complete accepted source

`LAMBDA_SPATIOTEMPORAL_EXECUTION_RECEIPT.json`
Structured final receipt and theorem ledger

`LEAN_WEB_MESSAGES.txt`
Complete final captured messages

`INITIAL_SOURCE_DIAGNOSTICS.txt`
Original-source diagnostic history; not PASS

`REPRODUCE_LEAN_WEB.url`
Captured source-bearing editor URL

`VERIFY_README.txt`
Verification and reproduction instructions

The prior printed SHA-256 value and pending-status attachments have been removed. The receipt uses direct source comparison and preserved source bytes. No self-referential digest of the finished PDF forms part of this edition.

Appendix C. References and corpus relation

- [1] Lu Semita. Language as a Technology for the Faithful Description of Experiential Reality: Faith, Definition, Identity, and the Pharmakon of the Descriptive Frame. EmergenceByDesign, Version 3.1, 8 September 2026.
- [2] Lu Semita. Constructive Truth-State, Semantic Time, and Adjudicative Will. EmergenceByDesign Lab Notes, Version 1.0, 8 September 2026.
- [3] Lu Semita. Increment, Registry, and Measure: Spatial-Temporal Co-Definition, Cyclic Return, Frequency, Scale, Numeral Bases, and Descriptive Bridges. EmergenceByDesign, August 2026.
- [4] Lu Semita. Expression of the Dimensional Ladder as Dual-Framed Description According to the Lambda Principle of Irreducibility. Verified V2 with integrated Lean receipt, 9 September 2026.
- [5] Lu Semita. Deflation of Pharmakon-Based Delusion: A Formal Example of Undisclosed Meaning Change Under One Public Label. Lean receipt edition, 9 September 2026.
- [6] Lu Semita. Before the Proof: Claim Decomposition, Constructive Certification, and the Separation of Formal Result from Interpretive Meaning. EmergenceByDesign, Version 1, August 2026.
- [7] Lu Semita. Nested Transfer and Scale-Free Spectra. EmergenceByDesign Lab Notes, September 2026.
- [8] Lu Semita. Zero Is Not Nothing: Boundary-Conditioned Vacuum Readouts, the Casimir Effect, and the Registry Errors of Zero-Point Engineering. EmergenceByDesign, September 2026.
- [9] Lean Project. The Lean Language Reference, “Elaboration and Compilation - The Kernel,” and Lean 4.33.0 release notes (10 August 2026). <https://lean-lang.org/doc/reference/latest/>
- [10] Stanford Encyclopedia of Philosophy. “Compositionality.” First published 2004; substantive revision 3 November 2025. <https://plato.stanford.edu/entries/compositionality/>

Corpus relation

This paper does not replace the companion works. Before the Proof supplies the general claim-routing architecture; the Dimensional Ladder supplies an operational and previously machine-checked projection/transport exemplar; Semantic Deflation supplies the machine-checked public-label example; Constructive Truth-State supplies the established definition of residual will and semantic-time architecture; Increment, Registry, and Measure supplies the operational space-time co-definition and causal-order caution; Language as a Technology supplies the inherited-interface, compression, pharmakon, and AI-as-contrast-instrument readings; Nested Transfer supplies the self/other perturbation seam, finite stopping discipline, and non-exterior comparison motif; Zero Is Not Nothing supplies a typed substitution and exclusion discipline in a concrete physical controversy. The present contribution assembles those components around reference polarity and the No-First-Position result while making the Lean translation boundary explicit.